

Optimizing the agroforestry residual biomass supply chain: A disruptive tool for mitigating logistic costs and enhancing forest management

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ABSTRACT

In a world with different energy needs, and more focused on renewable energies, biomass makes a very positive contribution. As a way of not creating more pressure on the forest, reducing the combustible loads of the forest, and reducing forest fires, residual biomass becomes an interesting resource. However, logistical costs and lack of information are threats to this valorization. In this sense, the objective of this work is to propose a technological tool (concept and prototype) that allows in an easy way and with high usability to connect the various stakeholders. This study began with an analysis of the state of the art, firstly carrying out a systematic review of the literature, accompanied by a benchmarking that led to a consolidation of the concept in a first phase, which was then followed by the construction of a prototype using the Figma tool. Thus, this study resulted in a disruptive tool that could optimize the information flow between the stakeholders, and that could result in benefits for all of them. The study presents some insights that can be used in the success of future projects in this field.

1. Introduction

In a world increasingly interested in renewable energy, biomass is an excellent value in this chapter [1]. The pressure on the forest (to exploit raw materials) requires a different look at resources that are usually unused, such as residual biomass that can be an excellent alternative to the exploitation of virgin material, with the added advantage of reducing forest fuel [2,3]. Although it seems like a good scenario, the high logistical costs, combined with the high interdependence between the different phases of the supply chain (SC), together with a lack of information among the different stakeholders [3], make this resource not exploited, often leading to it being burned in the open air and causing rural fires [4,5].

Many works have been discussed in the literature about the optimization of residual biomass SC. These optimizations list different contributions such as the exploitation of different conversion technologies [6,7], different locations for the infrastructures [8], or even changes in the processes [9,10]. To try to valorize waste biomass, some

platforms have emerged intending to connect buyers and producers [11, 12]. Some platforms add optimization models to this link [13], however, lack of knowledge, combined with the complexity of the systems, means that this world is not yet known, burning the forest leftovers.

The world of agroforestry residual biomass still does not have much work done in its optimization, more specifically in the information channels, and it is in this gap that this work tries to contribute. Thus, this work proposes a disruptive tool that aims to connect all agroforestry residual biomass supply chain (ARBSC) stakeholders, optimizing it, and providing valuable information for the decision-making processes of the various stakeholders. This tool emerges as the consolidation, presenting the final requirements list and the technological prototype, of a concept already presented in the literature [4].

To better understand the gap and important inputs for the following phases of development, in the first phase, a systematic literature review was made, followed by a benchmarking that then led to the construction of the tool presented here. Section 2 details the methodology used in this study, section 3 presents the state of the art, section 4 presents the

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results, and sections 5 and 6, present the discussion and conclusion, respectively.

2. Materials and methods

2.1. The analysis of the current state of the art

The first phase of this study is analyzing the state of the art concerning the topic of residual biomass supply chain optimization. To achieve this, two searches were conducted, a Systematic Literature Review (SLR) and an applications benchmarking (AB). The SLR has the aim of perceiving what literature discusses about the theme and AB has the role of completing the literature, doing a more comprehensive and freer search, seeking, for example in non-scientific sources.

Concerning the SLR, the search was made in the database SCOPUS. The construction of the final formula took place interactively to specify and find the ideal pool of articles to be analyzed. Thus, Table 1 presents the various formulas tested and the consequent number of results generated by them.

After finding the search formula, an SLR, following the criteria of the PRISMA methodology [14] was conducted. As all results are in English, all were included. After that, the Conference paper, Book chapter, and Review were excluded due they are not very relevant to the study in question, which result in a pool of 23 articles. As the pool of results was few, a filter by abstract was not conducted, being all papers included in the content analysis phase.

Regarding the AB with the aim of better understanding the existing gap and compare the purpose solution with the remaining ones. To achieve this purpose, terms such as “platform for the valorization of residual biomass”, “platform for optimization of residual biomass” and others were searched in the Google search engine.

2.2. The construction of the tool

In the process of creation disruptive tools to fill some gap, two big themes must be covered. The first part is the creation of the concept. Here it is important to perceive the various stakeholders (actors) and perceive the purpose of the tool. The other topic that should be developed is the construction of the tool prototype.

The first part was made based on the extension and improvement of a concept previously discussed in the literature [4]. Thus, to solidify the concept already presented, after analyzing the state of the art, meetings were held with experts in the field with experience both in academia and in the practical context to consolidate the entire final concept.

To achieve the second purpose, the construction of the platform, the Figma tool was selected, to develop a horizontal prototype, to avoid those major developments in terms of programming are made and not used. The Figma tool (Professional version) was chosen for all the features it has, such as writing CSS code, which will allow you to capitalize on time in subsequent phases, and for the fact that it is widely accepted in the developer community. In this sense, the first step of the creation was the layout definition; then, the definition of the content by interface; after that the selection of the resources that could better meet content needs (e.g. define logic for grouping information or types of buttons); the last phase is looking to the entire resources and define components, that could be reused, to simplify the development process. This philosophy goes in line with Component-Based Development, as indicated by Ref. [15], who defines that this as an approach where the developer

creates components and reuses them. The components’ ability to be reused raises the system’s overall reliability.

The entire methodology of this study was summarized in Fig. 1.

3. The current state of the art

3.1. Literature review

3.1.1. The biomass and residual biomass supply chain (RBSC)

A biological material produced by lived or recently lived organisms defines what biomass is [16]. Biomass is a resource that could act as an energetic backup of solar energy [3] and being valuable in the renewable energy chapter [1], due to the possibility of reducing carbon emissions. Contrarily, characteristics such as low density [17], decrease interest in this resource.

The high capacity to regenerate [3], coupled with the possible reduction of pressure on forests (to exploit raw materials) [1] may be factors that make the exploitation of waste biomass “attractive”. In addition, this can be vital in reducing residual fires by reducing fuel loads [2,3], but it remains important to bear in mind that these resources are also important in terms of soil integrity [1].

As regards the RBSC, it comprises three main stages: collection, transportation, and delivery [3,18,19]. For the central transportation cycle, waste can be unloaded, stored, and reloaded, and this loop can be repeated several times [3]. This temporary storage can provide added value by reducing moisture content. A lot of waste is cut and left on site, which makes it impossible for trucks to collect it. In this sense, the work of Nunes et al. [9] studies the possibility of optimizing this transport, which occurs in the first phase of the RBSC, using animal work.

The characteristics of biomass and the supply chain make logistics costs high. This chain has the particularity of having a lot of dependence and intensive work. Allied to this, the lack of information channels means that there is little exchange of information between all stakeholders [3].

3.1.2. The optimization of residual biomass supply chain (RBSC)

Based on the problem mentioned in the previous section, optimization has been the subject of several studies adopting different approaches and different final objectives, trying to optimize using technologies, changes in processes, location studies, or transport networks.

Most of the works focus on the economic impacts of the RBSC. Tagliabue et al. [7] evaluate the energy production potential using the anaerobic digestion technique, proposing a decision support system. The work of Pulido-Ocegueda et al. [6] tries to evaluate the utilization of different technologies to use the residual biomass, they evaluate the results based on economic and social factors. The utilization of combined heat and power could decrease the storage costs in the recovery of woody biomass [2].

The multimodal transport for the residual biomass is significantly interesting for big distances [18]. Sakar et al. [20] state that disruptions affect the sustainability of RBSC and state also that multimodal transport could act positively in this problem. The work of Santibañez-Aguilar [21] tried to study, based on Geographic Information System (GIS) and mathematical formulation, a RBSC model that aims to reduce installations and consequently increase profit. Residual biomass can, in certain contexts, replace crops, as in the case of agricultural biogas due to the optimization of the supply chain, as visible in the work of [22].

A mathematical formulation that aims to provide information on what wood from the forest should be collected per year (predicting the waste that will be generated there), taking into account disturbances (being the greatest the rural fires), and concluding that residual wood can have strong impacts on the mitigation of raw material failures is proposed by Rijal et al. [1]. Replacing coal energy by energy from residual biomass, and testing scenarios for changes in the cutting process, such as the creation of compact bales is discussed by Rivera-Cadavid

Table 1

Keyword used in each iteration and total of publications obtained.

Keywords	Total of publications
Residual + Biomass	9708
Residual + Biomass + Supply + Chain	122
Optimization + Residual + Biomass + Supply + Chain	26

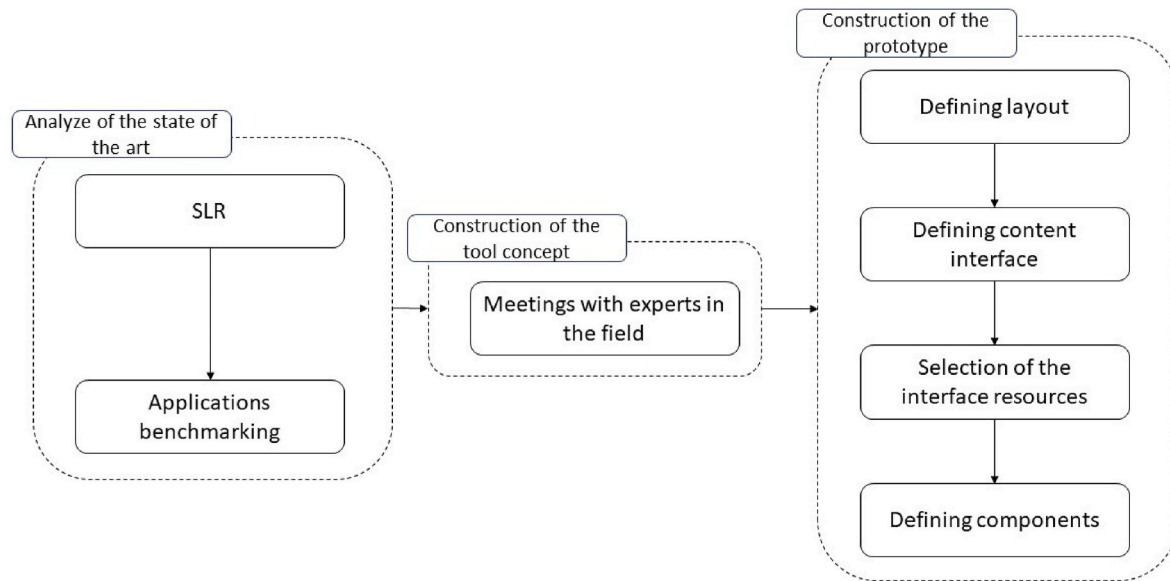


Fig. 1. Framework that summarizes the methodology conducted in this study.

[16].

A framework for SC sustainability that aims to minimize financial/environmental costs and increase associated professions was proposed by Ahmed & Sarkar [23]. They realized that the biggest contributor to the financial costs are the biorefineries and the environmental ones are the transports. The largest number of professions in this sector reside in rural areas.

Some works aim to study the optimal locations of production centers, logistic centers [24–26], transport models [27], or even both [8]. A formulation for the strategic planning for the use of pellets resulting from biomass from agricultural residues, which selects the residues, the optimal locations for the manufacture of the pellets, and even analyzes the distribution of the pellets is discussed by Martínez-Guido et al. [17]. Other work tries to investigate improvements in the supply chain by testing roasting and palletizing as a way to reduce total costs, techniques that are more relevant over long distances [10].

This study presents an optimum strategy for creating a resilient and sustainable biomass supply network based on the ambiguity of the bioenergy demand and the disruption of the biorefinery. Profit maximization is a sort of objective function [28]. A methodology to evaluate the techno-environmental suitability of biorefinery processes, considering the availability of biomass is proposed by Ref. [29]. Still in the evaluation chapter, Santibañez-Aguilar et al. [30] tried to formulate a criteria prioritization model that can be used to help decision-making.

3.2. Applications benchmarking

Concerning the applications benchmarking, some platforms exist with the aim of valorization the residual biomass, which are summarized in Table 2. These platforms have the main goal of linking biomass

producers with potential buyers or projects. **BiomassTrade** [11] primary goal is to open new opportunities for the companies that generate and process biomass. The BiomassTrade Platform gives biomass producers access to a brand-new market that adds value to currently discarded biomass leftovers and by-products. **BioEx Biomass Trading Platform** [12] allows connect participants in the Latvian biomass market, both suppliers and purchasers of biomass. Following registration, regular trade sessions, and open auctions where, like in the top commodities markets in the world, sellers and buyers select the price, transaction volume, supply time, or register collateral agreements, are organized. **forSCOPE** [13] is another platform that could allow the selling and buying of agroforestry leftovers, the users register the localization and choose the specie, type, amount of inert, quantity in tons, and humidity content. This platform has also a component of optimization, however, the user has the possibility of choosing the hyper-parameters for the algorithm. Within the theme of valuing biomass, there are applications with different purposes such as **ROSE-WOOD4.0** [31] which aims to encourage the sharing of information and practices on this subject, or **Bioraise** [32] which assesses prospective and available agricultural and forestry residual biomass resources, as well as their extraction and logistical costs, in the southern Mediterranean region. It is accessible online.

4. Results

4.1. The proposed tool: concept

One of the big challenges that agroforestry biomass faces in the energy recovery processes is the logistic costs. In this scenario, it is important to find mechanisms that allow a mitigation/decrease so that

Table 2
Overview of the most relevant key biomass trading and valorization platforms [11–13].

Feature/ Platform	BiomassTrade [11]	BioEx Biomass Trading Platform [12]	forSCOPE [13]
Primary Objective	Open opportunities for companies that generate/process biomass.	Connect Latvian biomass market participants both suppliers and purchasers.	Selling/buying of agroforestry leftovers
Access	Gives biomass producers access to new markets.	Following registration, regular trade sessions and open auctions are organized	Users register localization, choose specie, type, amount of inert, quantity in tons, and humidity content. Includes component of optimization.
Special Features	Adds value to discarded biomass leftovers and by-products.	Sellers and buyers select price, transaction volume, supply time, or register collateral agreements.	Users can choose the hyper-parameters for the optimization algorithm

forest management and consequent reduction of the fuel load become more “desirable” practices and the problem of fires be solved based on prevention and not reaction. One way that could act in this thinking line is creating mechanisms that could valorize the leftovers. For this reason, this work presents a disruptive tool that could aid the entire agroforestry residual biomass supply chain (ARBSC).

4.1.1. The ARBSC actors

According to the literature, the ARBSC comprises three big phases, the agricultural and forestry operations, which will generate biomass material, the transportation phase, and finally, the unloading and consequently storage. This loop could occur at various times; however, the phases are always the same.

Based on this theory, it is noticeable that ARBSC comprises three different profiles of participants and for that reason, the proposed tool will have three different profiles of users, the actors. Thus, the three actors of the system are: the **producer**, the **transporter**, and the **receiver**.

- The first actor, **producer**, corresponds to a profile of a user who generates agroforestry residual biomass (leftovers) from different activities. The main expected contribution of this platform to this profile is the collection and consequent monetary recovery of leftovers.
- The second actor, the **transporter**, corresponds to a player that makes the collection of the leftovers and consequent delivery to the final consumer. The major expected contribution of this tool to this actor is to have a detailed route that allows diluting the maximum transportation costs (time and fuel) to make the transportation of this waste feasible.
- The third actor, the **receiver**, corresponds to the player that collects the biomass. The main expected contribution of this tool to this profile is the
- possibility of having low-cost materials to avoid buying more expensive materials.

Mentioned the participants of ARBSC and establishing the big contribution of the platform for each one, it is easy to identify a need in the system, an entity that could be responsible for the entire management of the platform. As this theme is so important to the political power for all the reasons that have already been discussed, they have a special interest in these questions and for this, could be the key player in the management of the platform, so the platform will include a fourth actor, the **member of the political authority**.

4.1.2. The platform requirements by actor

Aiming to create a tool that fulfill the needs of the actors above

Table 3
Specific requirements by actor.

Actor	Requirements
Producer	Communicate agroforestry leftovers; Visualize financial transaction data;
Transporter	Visualize the collection route; Visualize the collection route history; Visualize indicators; Communicate the collection availability window; Communicate the transport type (capacity, fuel usage ...); Visualize the collected points and their leftovers;
Receiver	Insert what type of leftovers s/he accepts, Communicate the discharge availability window; Visualize the “loads” history; Insert financial transaction data;
Member of political authority	Visualize indicators about the region's status; Register new members in the platform; Validate pre-register made by possible users;

mentioned, a set of requirements for each one was defined and presented in [Table 3](#).

Concerning the **producer**, although this profile just has two requirements, the first one, “Communicate agroforestry leftovers”, constitutes a challenge in this context because there is a need to create a standard intuitive process to facilitate the insertion of information. For this reason, the platform should contemplate a list of possible leftovers for the user to choose from. This mechanism will decrease the errors and will facilitate features of other actors, because leftovers from a given list allow, for example, algorithms to “know” a lot of information about the leftover. If this information was recorded with a free text field, a little writing error could take power away from the output of the application. Other important information in the communication of the requirements is the location of them, and in this way, this platform uses the ways of reporting a location already known from other apps like Google Maps (writing the address, entering the coordinates, marking on the map, or obtain the location of the device).

Regarding the **transporter**, the major challenge is to provide the maximum information to the system to allow to produce the most “desirable” routes. In this sense, the interface should be intuitive and with closed fields to avoid errors in the algorithm. This information will also impact the entire supply chain affecting the remaining actors.

Although this system comprises various types of information, this could be combined with external information, creating valuable indicators that can be an added value for the action of **members of political authority** which thus has a double benefit (joint with the reduction of fires). This could “minimize” the management cost and increase the interest in this platform.

4.2. The proposed tool: prototype

The big aim of this tool is to guarantee the flow of information throughout the various actors of the ARBSC, always providing tools that could aid in the decision-making process. The ARBSC has three main actors, the **producer**, the **transporter**, and the **receiver**. Besides these three participants, the **member of the political authority** has an important role in the system because has the responsibility of managing the entire operation of the platform.

Regarding the first actor, the **producer**, when s/he starts the platform, s/he has access to an interface ([Fig. 2. \(a\)](#)) where s/he can consult his/her communication history in the system (at the bottom) and has access to a button (at the top) that redirects him/her to the insertion of a new leftovers record.

After entering the part of inserting new leftovers ([Fig. 2. \(b\)](#)), s/he has access to a bulleted list of leftovers, where s/he can communicate which leftovers s/he has. After choosing the type, the system redirects the user to [Fig. 2. \(c\)](#), where the user could indicate the volume of the load and whether it is already available (at insert insertion time or later). Finally, the last step of reporting leftovers is to indicate where they are ([Fig. 1. \(d\)](#)), and this can be done by manually writing the address, entering the coordinates, marking the map, or choosing to obtain the location of the device.

The second actor of this system, the **transporter**, has access, in the first phase, to an interface ([Fig. 3. \(a\)](#)) that provides him/her with some basic indicators, as well as the route history and the next route. Below the header, by pressing the second button, s/he has access to a time grid ([Fig. 3. \(b\)](#)), where s/he can enter what availability s/he has to make the transport and can also see which windows already have services allocated. The last tab has two big components, the first one, [Fig. 3. \(c\)](#), shows the collection points where the transporter has already passed, and the transporter can see if any of them has leftovers not yet allocated to routes. Also in this tab, the types of transport that the hauler has are visible ([Fig. 3. \(d\)](#)), indicating the fuel consumption and capacity, which will work as a basis for the algorithm that will give the routes.

The last user of the ARBSC is the **receiver**. This profile, like the others already mentioned, has a stack of indicators as its first interface



Fig. 2. Interfaces for **producer** actor: (a) main interface; (b) leftover type selector; (c) volume/availability conditions; (d) leftovers location.

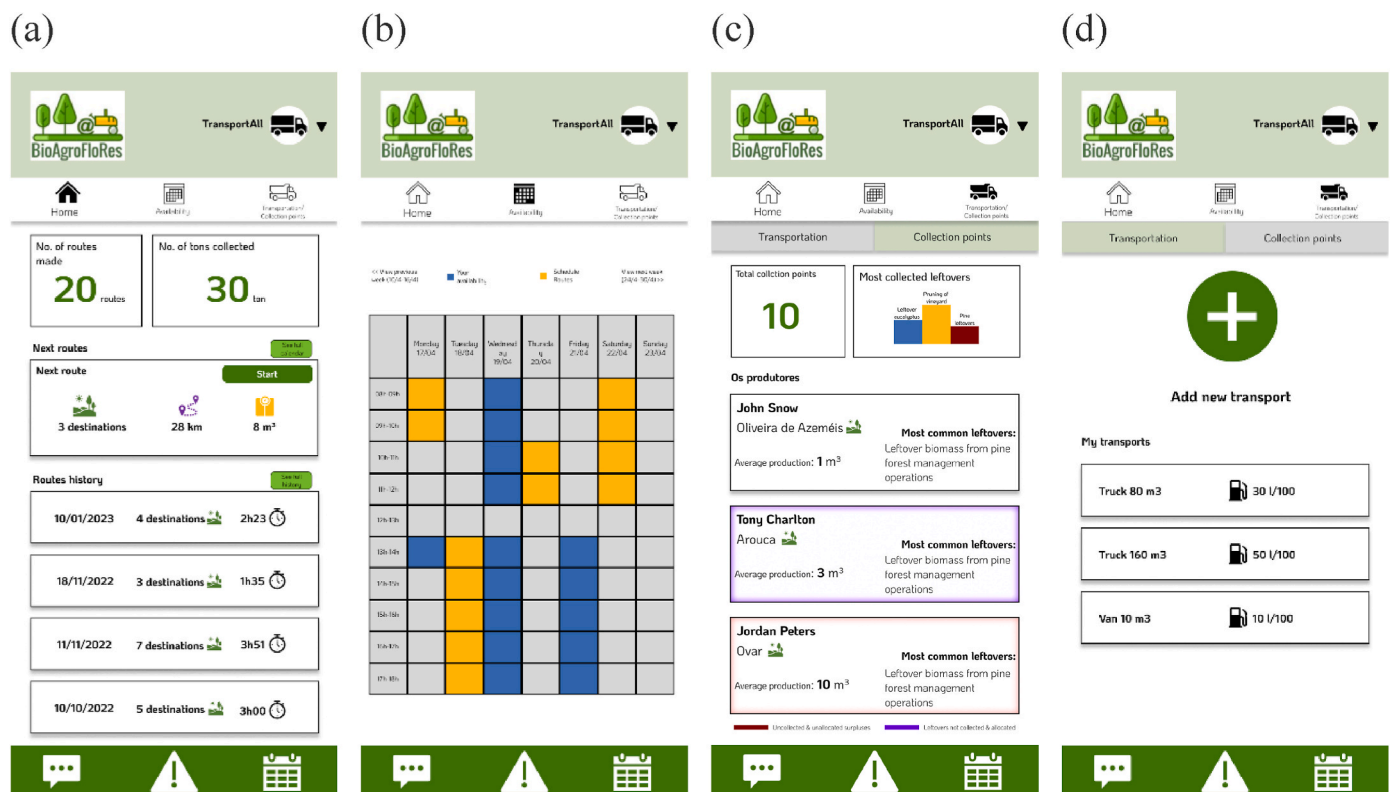


Fig. 3. Interfaces for **transporter** actor: (a) main interface; (b) transporter timetable; (c) collection points; (d) types of transportation.

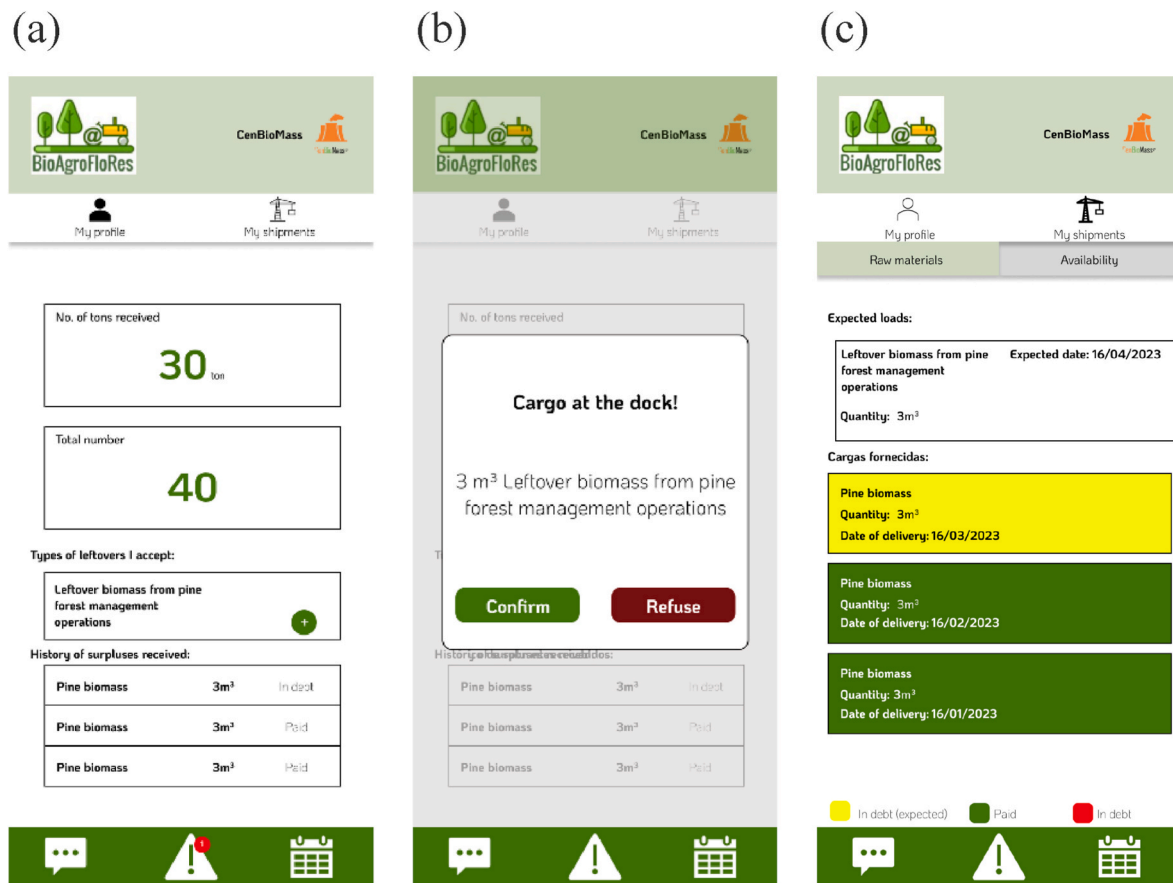


Fig. 4. Interfaces for receiver actor: (a) main interface; (b) dock cargo notification; (c) financial transaction interface.

(Fig. 4. (a)). Whenever the transporter arrives at an unloading site, the receiver receives a notification to check if s/he has received the load in the system (Fig. 4. (b)). After closing the cycle, the load moves to a cargo history where the receiver must manage all financial transactions (Fig. 4. (c)). In the case of this actor, as in the previous one, there will also have a window to communicate the availability for the reception of the load and at the same time to realize it already has loads allocated in that time window.

The last player presented here is **member of the political authority**, who has an important role in system management. The member of the political authority has two important features in the system, **firstly**, the user management (Fig. 5. (a); (b)), where it validates new users and accesses information from existing ones, **secondly**, the access to relevant information (Fig. 5. (c)), the result of data from the system and from external systems, about the state of the municipality that can help the decision-making of the political power on this subject.

5. Discussion

The use of agroforestry biomass is an important issue that can add a lot of value to society at various levels, such as energy use and fuel reduction in forests [2,3]. Allied with this, there is a growing concern about rural fires that can be helped with the use of these resources. Analyzing the literature, it is clear that there are many problems in terms of the management of residual biomass chains that make this use not often done. One of the major challenges lies in the logistical costs [3]. The exploitation made concerning the optimization of the residual biomass supply chain is made in different ways, such as the utilization of different technologies [6,7], studies at the level of means of transport [18,20], the planning of plant locations [8], algorithms to plan material collection and changes to processes [9,10].

There is a serious problem identified in terms of the lack of information among the various stakeholders [3] that the various results in the literature regarding the optimization of this chain do not explore. In this sense, the proposed solution aims to respond to the problem identified above and increase the literature proposing a different approach. The proposed concept aims to reach the three players in the three major stages of the residual biomass supply chain [3,18,19]. For this reason, these three actors were the three profiles that the platform contemplate. As the rural fires impact a whole society, the member of the political authority was also created as an actor. This profile will act as a manager of the entire platform.

Although the literature does not focus on this kind of platform, some applications have been developed with the aim of valorize the residual biomass. Comparing the proposed solution with the solutions described in AB, BiomassTrade [11], and BioEx [12], it is evident that these solutions seem to differ in that the solution discussed in this work includes a strong component in helping transport operators to make decisions (actor: Transporter), optimizing their routes. Another vital difference is to put the center of this technology in combat of rural fires instead of trying to create a marketplace, which could have strong participation by the political member because they have economic benefits, which can be useful in dissemination. In the case of the Portuguese context, this platform must present the Portuguese language, this work presents in English to facilitate the reader's comprehension, because the literacy of many players does not include knowledge in English. The BioEx system is conceived for the Latvian context which makes its use unfeasible in the Portuguese context.

The solution forSCOPE [13] is close to what was done here, however, it is a tricky solution that requires a lot of information that users will not be able to answer, such as the number of aggregates, the moisture content, or the amount in tons. Allied with this, the type of



Fig. 5. Interfaces for members of the political authority actor: (a) interface to validate new users; (b) users information interface; (c) General indicators dashboard (interfaces presented in Portuguese).

surplus includes wooden rollaway, which in the Portuguese context is seen as raw material, so this application can be seen as an application that aims at something beyond residual biomass. Furthermore, some hyper-parameters in the optimization algorithm can be difficult to estimate by people who are not knowledgeable in the field of operational investigation. In this way, the poor usability and complexity of the system can be threats to the system and in this sense, this proposed platform aims to solve these two points.

Many of these cases, such as **forSCOPE**, are developed in areas of projects that finally end and then there is no continuity in use and future developments of the system. This application tries to involve the actor member of the political authority to try to encourage them to use it, often with incentives or measures, in this sense, the application revenues (with prevention of rural fires) can generate returns to invest in the maintenance and evolution of the system until it becomes common practice throughout the country.

It is noted that at the optimization level, the financial costs [21,22] are the ones that weigh more in the objective functions of the optimization models, being also the environmental issues relevant to these objectives [23]. In this sense, the proposed platform goes in line with these results by trying to optimize the entire chain allowing receivers to have cheaper material, at the level of optimization algorithms also trying to propose interesting benefits to transporters and finally, those who sometimes do not see their forestry waste valued, can also derive some gain from this system. At the environmental level, two major gains are evident, the first in terms of reducing CO₂ emissions from rural fires (which are thought to be minimized) and then by the algorithm that will be the basis of the construction of the routes. Still in terms of the construction of the routes, elements such as the time that the leftovers are left outdoors or the inclusion of techniques such as palletization can also

be added value to both economic and environmental objectives. Regarding the last component of sustainability, the social pillar, some works also try to study the impact there [6,13]. In this pillar of sustainability, the reduction of fires can be a social enhancer, as it does not endanger lives or people's homes.

6. Conclusion

This work has as a contribution a proposal for a platform that intends to mitigate a gap identified in the literature. The purpose of this platform is to add a communication channel that still allows adding relevant information to decision-making. At the theoretical level, this work presents a proposal for a theoretical concept for a tool, identifying the intervenient and the requirements that could meet their needs. Also, a state-of-the-art that complements not only the contributions from scientific bases but also adds research in non-scientific sources. This could be an important contribution to future developments in this field. At a practical level, the platform will allow for better communication between the various actors in the agroforestry biomass supply chain. In addition, the discussion presents perspectives that can lead to the success of these models (which already exist but are not used) because, besides the creation of tools, it is important to perceive certain factors that could be crucial in their success. As a limitation, it is possible to point out the possible lack of greater listening to the various players in the chain. Future work can incorporate a route optimization model, which can be derived from the literature, or developed based on real data that can be collected. In addition, validation tests can be developed to collect opinions from field players that allow a stronger validation of the proposed solution.

Author contributions

All authors contributed equally to the article. All authors have read and agreed to the published version of the manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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